TAMIL NADU STATE BOARD - STANDARD 12 PHYSICS VOLUME 2

CHAPTER 9: ATOMIC AND NUCLEAR PHYSICS

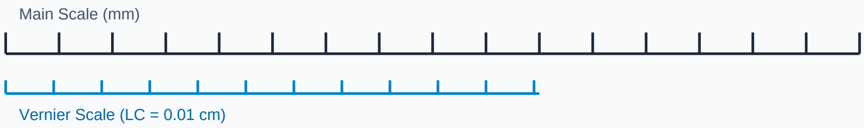
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Physics Volume 2
Chapter Name:	Atomic and Nuclear Physics

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division



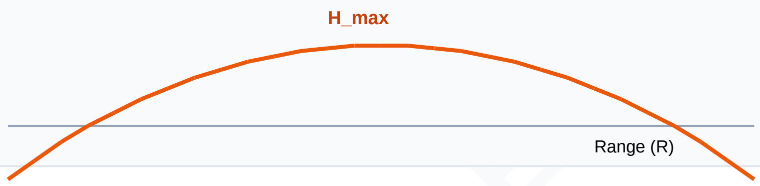
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Impact Parameter	The perpendicular distance of the velocity vector of an alpha particle from the center of the target nucleus.
Mass Defect	The difference between the sum of masses of individual nucleons and the actual mass of the nucleus.
Half-Life Period	The time required for half the number of radioactive nuclei in a sample to decay.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

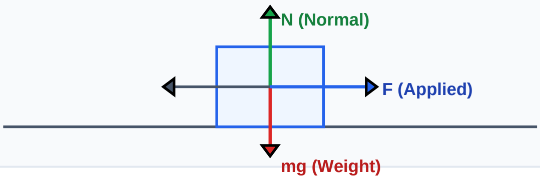
Kinematic Projectile Trajectory



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Bohr Orbit Radius: $r_n = a_0 * n^2$ | Energy: $E_n = -13.6 / n^2$ (eV)
- Radioactive Decay Law: $N(t) = N_0 * e^{(-\lambda * t)}$ | Half-life $T_{0.5} = 0.693 / \lambda$
- Binding Energy: $BE = [Z*m_p + (A-Z)*m_n - M_{nuc}] * c^2$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

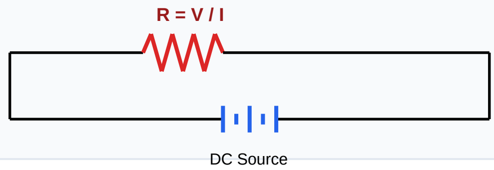
Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: If half-life of a material is 5 days, how much of a 10g sample remains after 10 days?

Step-by-step Solution:
Number of half-lives $n = 10 / 5 = 2$.
Remaining mass $N = N_0 * (0.5)^n = 10 * (0.5)^2 = 10 * 0.25 = 2.5$ grams.

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

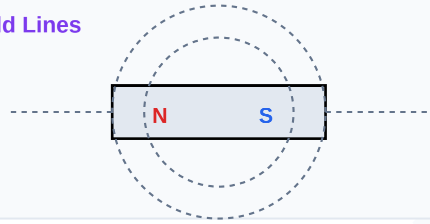
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Derive the expression for the radius of the nth Bohr orbit of a hydrogen atom.

Analytical Solution Strategy:

Electrostatic force balance: m*v^2/r = e^2 / (4*pi*eps_0*r^2) => m*v^2 = e^2 / (4*pi*eps_0*r).
Bohr quantization: m*v*r = n*h / 2*pi => v = n*h / (2*pi*m*r).
Substitute v: m * [n^2*h^2 / (4*pi^2*m^2*r^2)] = e^2 / (4*pi*eps_0*r).
Simplify to get r_n = (n^2 * h^2 * eps_0) / (pi * m * e^2).

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Bohr's three postulates of atomic model.

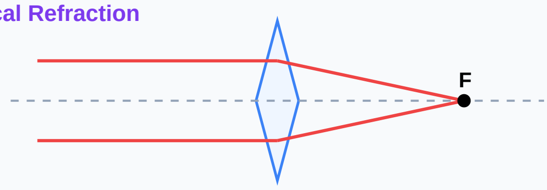
Q2 (2-Mark): Write down Rydberg formula for spectral series of hydrogen.

Q3 (2-Mark): Define nuclear fission and fusion with example reactions.

Q4 (5-Mark): Explain binding energy per nucleon curve significance.

Q5 (5-Mark): State radioactive displacement laws (alpha, beta decay).

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Energy Grid

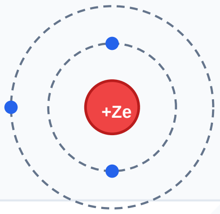
Nuclear power stations generate steady steam volumes using uranium fission reactions.

Application Case: Medicine

Radioactive isotopes like Iodine-131 act as tracing flags to localize thyroid tumors.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Sinusoidal AC Voltage Waveform